A-A-53235
29 September 1987

COMMERCIAL ITEM DESCRIPTION

WIRE, NONELECTRICAL (Chromium-Cobalt Alloy)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a Wire, Chromium-Cobalt Alloy, 18 Gage.

Salient Characteristics:

The wire shall be of Chromium-Cobalt, for use in fabricating clasps and partial dentures.

Chemical composition. The wire shall be Chromium-Cobalt Alloy and the chemical composition shall be as follows as shown in Table I:

TABLE I. Chemical Requirements

Element	Composition %
Cobalt	48-52
Chromium	17-22
Tungsten	12-15
Nickel	8-11
Manganese	1-2
Iron	1-2

Dimensions. The wire shall be round in shape, with a diameter of 18 gage 0.0400 ± 0.0008 inch. The wire shall be a minimum of 1 foot in length.

AMSC/NA FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Finish. The wire shall be furnished in bright-annealed, and shall have a smooth, polished finish.

Mechanical Properties. The minimum mechanical properties for the wire shall be as follows: Tensile strength, 145,000 PSI; Yield strength, 125,000 PSI, and Elongation, 12%.

The wire shall be physically and chemically compatible to cast and weld to ticonium premium 100 alloy.

Corrosion test. The wire shall show no corrosion when boil tested as specified herein.

Workmanship. The wire shall be of uniform quality and temper, smooth, clean, and bright. Wire shall be free from splits, kinks, tarnish, or other defects which may detract from the appearance or impair the serviceability of the wire.

Unit. Tube (TU). One tube containing five, one foot length of wire, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one wire. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Tests.

Resistance to boil: Before testing, the wires shall be scrubbed with soap and water and rinsed thoroughly in distilled water, and dried. After being cleaned the wire shall be boiled in distilled water in a glass container for 30 minutes and allowed to remain submerged for an additional 24 hours. The specimens shall then be removed from the water and examined for corrosion.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers, own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	ITEM IDENTIFICATION
6520-01-041-8156	WIRE, NONELECTRICAL, CHROMIUM-COBALT ALLOY, 18 GAGE, 1 FT length, 5s.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PGC 48998

Project No. 6520-2022

Location: NCR Disk #4

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INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53235	WIRE, NON-ELECTRICAL (CHROMIUM-COBALT ALLOY)

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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